

ASIMOV v1

ENGINEERING BUNDLE — Concept Phase

Architecture · Requirements · DFMEA · Test Plan · Safety Case · 2026-05-03 · github.com/jherrodthomas/asimov-v1

What's in this bundle. A consolidated view of the concept-phase engineering package for the Asimov v1 open-source humanoid robot. Detailed evidence lives in the companion workbooks (Safety Case · DFMEA · Requirements · Test Plan); this PDF is intended for stakeholder review and decision-making. Per the Safety Case, deployment is provisionally accepted only for OE-1 (research lab, supervised, tethered) at this revision.

Document control

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Item	Asimov v1 humanoid robot — github.com/jherrodthomas/asimov-v1
Linked artefacts (xlsx)	Asimov_v1_Safety_Case.xlsx · Asimov_v1_DFMEA.xlsx · Asimov_v1_Requirements.xlsx · Asimov_v1_Test_Plan.xlsx
Linked artefacts (docx)	Asimov_v1_Safety_Case_Report.docx · Asimov_v1_Architecture.docx
Diagram pack	Asimov_v1_Diagrams/01..05 .svg + .png

Headline numbers

Hazards	Safety funcs	Requirements	DFMEA modes	High-AP	Test cases	Open gaps
32	25	120	32	19	139	28

1. Architecture (5 views)

The architecture is presented through five complementary views — system context, functional block, safety-function allocation, CAN bus topology, and mode FSM — corresponding to ISO/IEC/IEEE 42010:2022. Each view has been derived from the upstream repository and the Asimov manual; assumptions and gaps are tracked in the Safety Case Gap Register.

Asimov v1 — System Boundary & Context (ISO 12100 §5 / ISO 13482 §5)

Item under assessment, actors, environment, and external interfaces

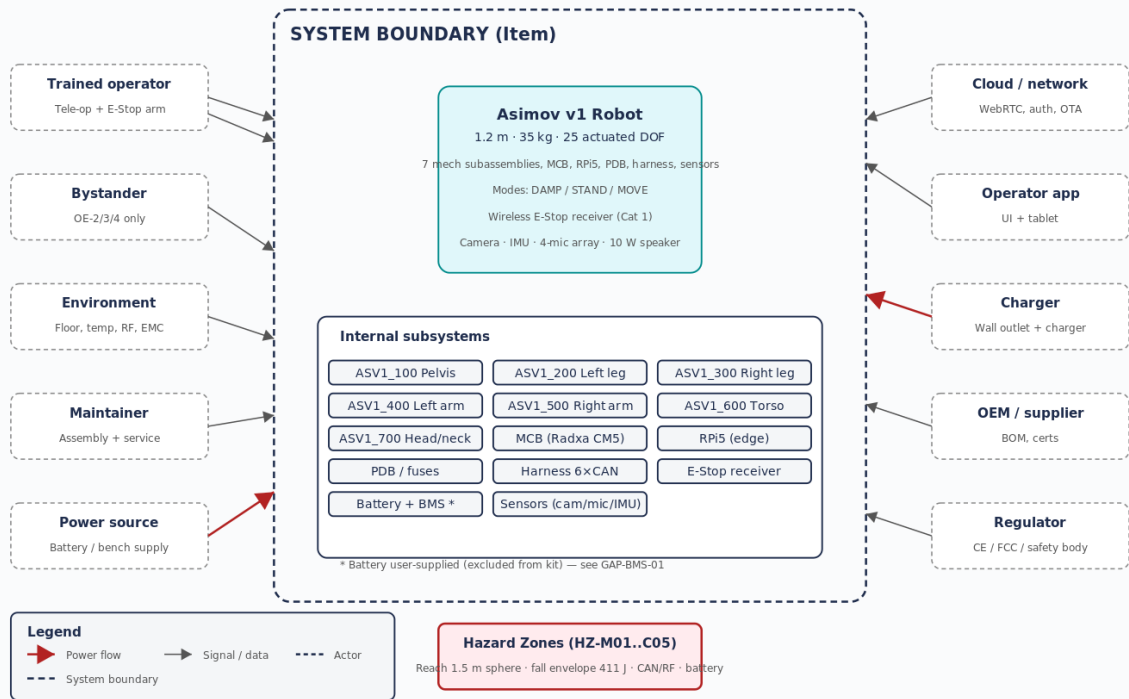


Figure 1 — System boundary & context (ISO 12100 §5 / ISO 13482 §5).

Asimov v1 — Functional Block Diagram

Subsystems, signal & power flows, six CAN buses

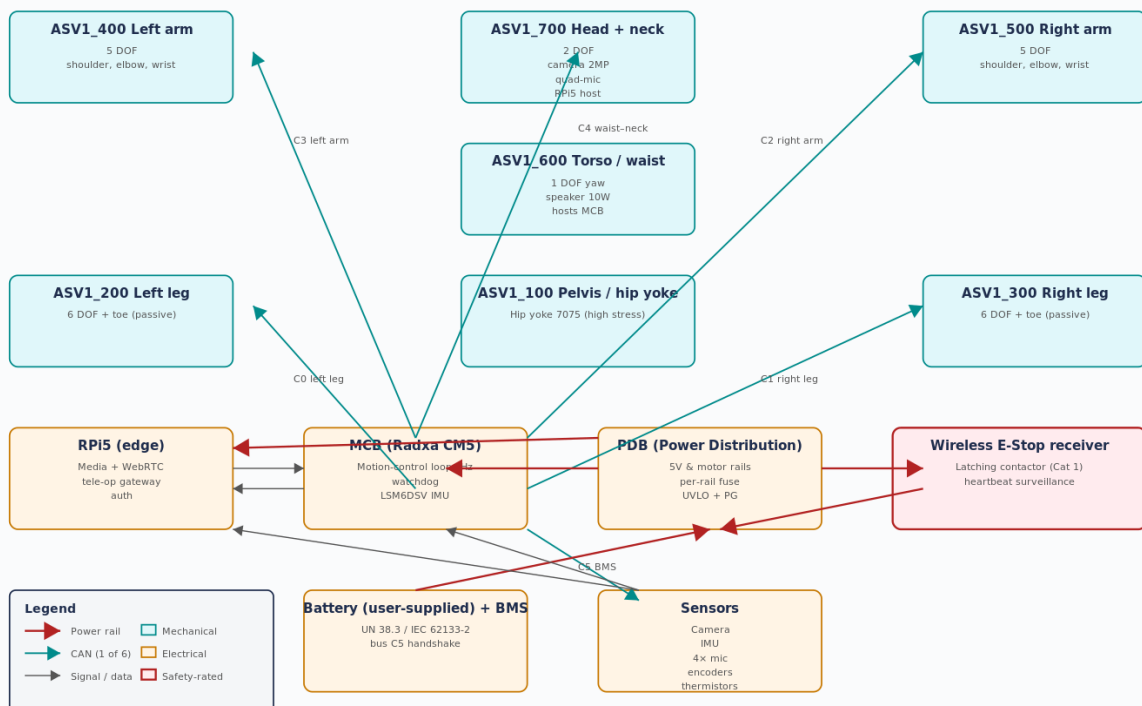


Figure 2 — Functional block diagram with 6 CAN buses, power, signal flows.

Asimov v1 — Safety-Function Allocation Map (25 SFs · ISO 13849-1 PL · ISO 13482 §6)

Each safety function is allocated to hardware, software, or procedural enforcement

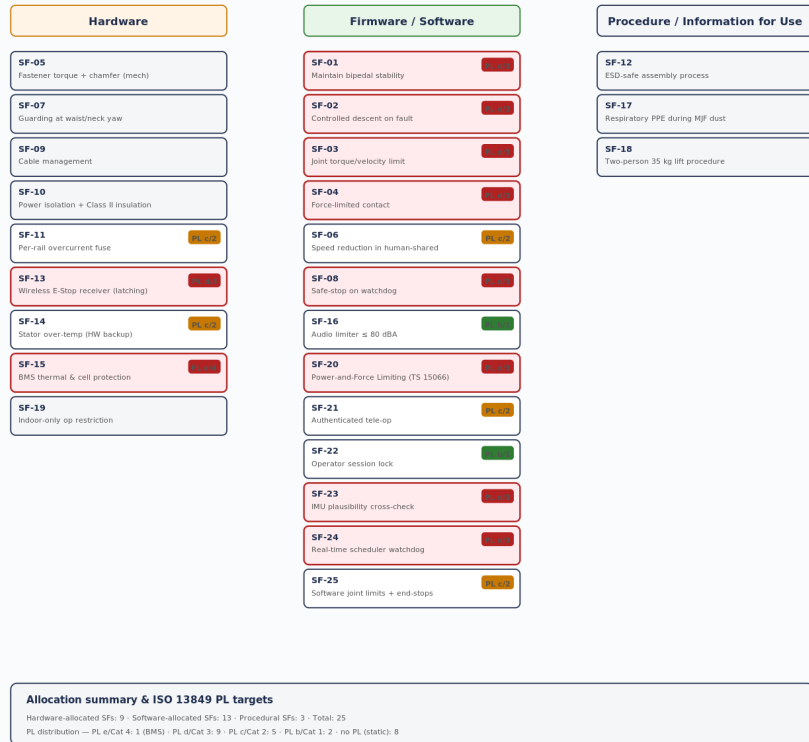


Figure 3 — 25 safety functions allocated to HW (9), SW (13), procedural (3) with PL targets.

Asimov v1 — CAN Bus Topology (6 buses · 5×1 Mbps + 1×500 kbps)

Each bus terminates 120 Ω at both ends; bus-off $\rightarrow$ DAMP within 100 ms (SF-08)

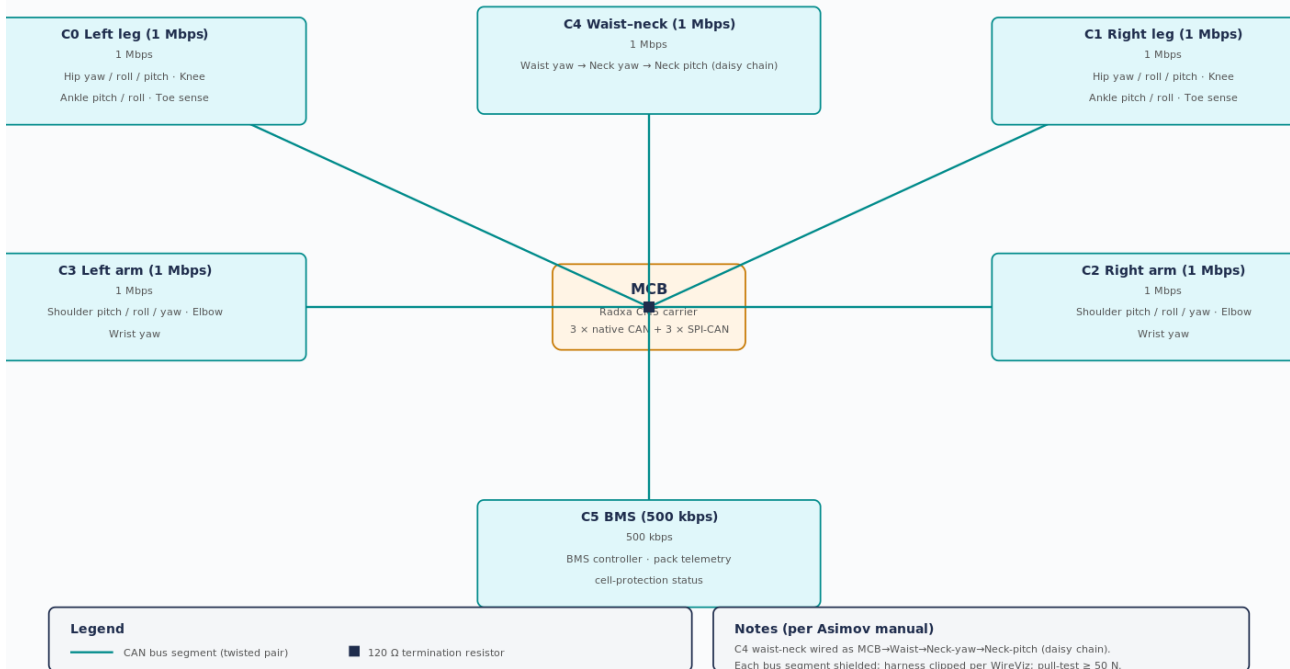


Figure 4 — CAN bus topology: C0..C5 with 120 Ω termination.

Asimov v1 — Mode FSM (DAMP / STAND / MOVE) with safety gating

Watchdog and E-Stop gate every transition; default-on-fault is DAMP

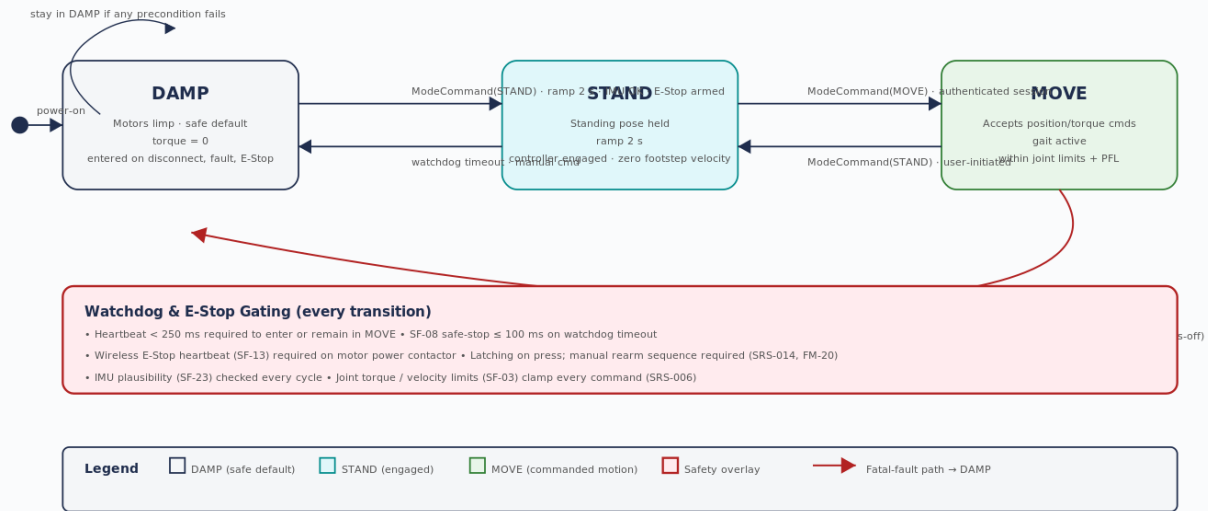


Figure 5 — Mode FSM (DAMP / STAND / MOVE) with watchdog and E-Stop gating.

2. Requirements summary

120 requirements across SyRS (35), SRS (45), and HwRS (40). 63 are safety-classified and 9 are cyber-security. Each is traceable to ≥ 1 safety function, hazard, FMEA failure mode, gap, and test case in the companion workbook ([Asimov_v1_Requirements.xlsx](#)).

Top 12 mandatory safety requirements

REQ-ID	Type	Statement (paraphrased)	SF	Tests
SYRS-002	SAFE	Controlled descent on fault — ≤ 200 ms detection-to-stable	SF-02	TC-S-002
SYRS-003	SAFE	Wireless E-Stop ≤ 500 ms; fail-safe on link loss	SF-13	TC-S-003,4
SYRS-004	SAFE	Per-joint torque limit, software + hardware	SF-03	TC-U-001
SYRS-005	SAFE	Per-joint velocity limit	SF-03	TC-U-002
SYRS-006	SAFE	PFL per ISO/TS 15066 in cobot mode	SF-20	TC-S-006
SYRS-007	SAFE	Watchdog DAMP on disconnect ≤ 350 ms	SF-08	TC-S-007
SYRS-013	SAFE	Battery UN 38.3 / IEC 62133-2 + BMS	SF-15	TC-S-010
SYRS-014	SAFE	BMS interlock — no motor power without handshake	SF-15	TC-I-002
SYRS-016	SAFE	IMU plausibility cross-check	SF-23	TC-U-007
SYRS-021	SAFE	Tip-over under 50 N lateral	—	TC-S-013
SYRS-029	SAFE	Guarding ≤ 4 mm at rotational interfaces	SF-07	TC-U-011
HwRS-019	SAFE	E-Stop latching contactor on link loss > 1 s	SF-13	TC-S-042

3. DFMEA — top High-AP items

32 failure modes scored AIAG-VDA 2019: **19 High** · 11 Medium · 2 Low. Every High-AP item has a recommended action linked to a gap in the Safety Case. The full DFMEA lives in **Asimov_v1_DFMEA.xlsx**.

FM	Description	S	O	D	AP	Mitigation	Linked gap
FM-19	Wireless E-Stop link loss without fail-safe	10	3	4	H	E-Stop arch	GAP-13482-04
FM-21	Li-ion cell internal short → thermal runaway	10	3	5	H	BMS / pack	GAP-BMS-01
FM-22	BMS missing or non-compliant	10	5	10	H	Kit policy + interlock	GAP-BMS-01
FM-01	Hip-yoke fatigue crack at high-stress fillet	9	3	7	H	FEA + lifecycle test	GAP-13482-03
FM-02	Substituted material — 6061 instead of 7075	9	5	9	H	Mill-cert gate	GAP-9001-03
FM-05	Actuator runaway (full torque)	9	3	4	H	Cat 3 torque-monitor	GAP-13849-01
FM-11	CAN leg bus → bus-off	9	3	3	H	Bus-off → DAMP, redundancy	GAP-13849-01
FM-13	Watchdog not armed	9	3	6	H	CI test + ISO 13849-2	GAP-13849-03
FM-15	Tele-op auth bypass / weak token	9	3	7	H	TLS+token+audit	GAP-CYB-01
FM-16	Fuse rated too high / missing	9	2	5	H	Locked BOM + boot test	—
FM-17	XT30 reverse polarity / hot-plug	8	3	5	H	Reverse-polarity protection	—
FM-28	Locomotion policy edge case	9	5	7	H	Sim-to-real coverage matrix	GAP-LP-01

4. Test plan summary (V-model)

139 test cases — Unit (83) · Integration (24) · System (32). All 25 safety functions are mapped to ≥ 1 ISO 13849-2 validation activity; ISO 13482 §7 V&V matrix is fully populated. Full plan in **Asimov_v1_TestPlan.xlsx**.

Top safety system tests (selected)

TC-ID	Title	SF	Method
TC-S-001	Bipedal walk 5 minutes (OE-1 lab)	SF-01	T,D
TC-S-002	Controlled descent on simulated comms loss	SF-02	T
TC-S-003	Wireless E-Stop response ≤ 500 ms	SF-13	T
TC-S-004	E-Stop link-loss fail-safe ≤ 1 s	SF-13	T
TC-S-005	Joint torque overload protection	SF-03	T
TC-S-006	ISO/TS 15066 PFL measurement campaign	SF-20	T
TC-S-010	BMS compliance gate — non-compliant pack rejected	SF-15	T
TC-S-013	Tip-over under 50 N lateral	—	T
TC-S-014	E-Stop link-loss soak (24 h)	SF-13	T
TC-S-020	Locomotion policy verification across 32-hazard catalogue	SF-01	T,A
TC-S-041	Battery thermal-runaway containment (UN 38.3 internal-short)	SF-15	T
TC-S-042	Wireless E-Stop architecture validation per ISO 13849-2 Table B.1	SF-13	T,A

5. Open gaps (top 12 of 28)

Closing these unblocks OE-1 unconditionally and OE-2 conditionally. OE-3/OE-4 require additional regulatory work.

GAP-ID	Standard / clause	Description	Severity
GAP-13482-04	ISO 13482 §5.16	Wireless E-Stop architecture, latency, link surveillance	VHIGH
GAP-10218-03	ISO 10218-1 §5.5.3	E-Stop architecture (Cat 0/1, dual-channel)	VHIGH
GAP-13849-02	ISO 13849-1 §7	MTTFD/DC/CCF figures absent — PL cannot be computed	VHIGH
GAP-13849-03	ISO 13849-2	Validation plan & test report missing	VHIGH
GAP-13482-01	ISO 13482 §5.7	Force/energy limits not declared	VHIGH
GAP-13482-05	ISO 13482 §6	Safety-related control-system implementation	VHIGH
GAP-13482-06	ISO 13482 §7	V&V plan and report not produced	VHIGH
GAP-13849-01	ISO 13849-1 §6	Cat 3 redundancy not implemented for 10 SFs	VHIGH
GAP-BMS-01	IEC 62133-2	Battery user-supplied — no compliance gate	VHIGH
GAP-LP-01	(this case)	Locomotion policy unreleased	VHIGH
GAP-12100-01	ISO 12100 §6.4	Adequacy-of-risk-reduction record absent	HIGH
GAP-DOC-01	(this case)	Manual /v1/safety, /v1/quickstart, /v1/bom return 404	HIGH

Recommendation. Provisional acceptance for OE-1 (research lab, supervised, tethered) only. Closing the top 12 gaps unlocks OE-1 unconditionally and OE-2 conditionally. Deployment to OE-3/OE-4 (home / public space) requires all 28 gaps plus regulatory engagement.

Companion artefacts. [Asimov_v1_Safety_Case.xlsx](#) (15-tab workbook) · [Asimov_v1_Safety_Case_Report.docx](#) (narrative) · [Asimov_v1_Safety_Case_Summary.pdf](#) (exec summary) · [Asimov_v1_DFMEA.xlsx](#) (12-tab AIAG-VDA 2019) · [Asimov_v1_Requirements.xlsx](#) (9 tabs, 120 reqs) · [Asimov_v1_Architecture.docx](#) (5 views) · [Asimov_v1_TestPlan.xlsx](#) (V-model, 139 cases) · [Asimov_v1_Diagrams/01..05 .svg](#) + [.png](#).